Metadata form :

Urban flooding historical database in HCMC

Layer name	Flood observations in HCMC
File name	
Thematic keywords	flood, flood observation, HCMC
Geographic keywords	Ho Chi Minh City, Vietnam, Southeast Asia
Summary	Point layer recording historical urban flood observations in HCMC. Each point represents one spatially localized flood event extracted from various sources (newspapers, reports, databases).
Author	Gintare RADZIUTE
Number of objects	
Number of attributes	
Collecting date	2026
Latest edit date	
Year	2026
Source (name and/or website)	Newspapers, scientific articles, reports, databases
Person(s) in charge of integrating data	Sébastien Hardy
Restrictions on use	
Geometry type	Point
Coordinate system	VN-2000 TM-3 105-45 (EPSG 9210)

General attribute structure

Description	What the field represents
Type	Data type (Integer, Text, Date…)
Unit	Unit of measurement (if numeric)
Source	Where the data comes from
Min / Max values	Range of values (if numeric)
Date	Date of last update or collection

Notes	Additional information or possible values

All attributes

Alias name : Obs_id

Description	Unique identifier for each localized flood observation.
Type	Text (str)
Unit	-
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	

Alias name : Date

Description	Date of the flood event.
Type	Date
Unit	YYYY/MM/DD
Source	Newspapers, Technical report
Min / Max values	-
Date	2026
Notes	

Alias name : Year

Description	Year extracted from the date field.
Type	Integer (Long)
Unit	Year
Source	Derived from Date field
Min / Max values	0 ; 2026

Date	2026
Notes	

Alias name : Location_name

Description	Name of the flooded location.
Type	Text (str)
Unit	-
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	Can refer to a street, intersection, ward, commune, school, airport, canal, airport or market area, etc.

Alias name : Location_type

Description	Type of location mentioned in the source.
Type	Text (str)
Unit	-
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	-

name : District_mentioned

Description	District mentioned in the source. Preserves the administrative reference used in the original document.
Type	Text (str)
Unit	-
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	

name : Province

Description	Province mentioned in the source (Before 2025 administrative reform)
Type	Text (str)
Unit	-
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	Possible values : Bình Dương or Bà Rịa-Vũng Tàu

Alias name : Ward/Phường

Description	Ward mentioned in the source.
Type	Text (str)
Unit	-
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	

Alias name : Commune/Xã

Description	Commune or xã mentioned in the source.
Type	Text (str)
Unit	-
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	This field is useful for observations located in peri-urban or formerly provincial areas.

Alias name : Current_admin_unit

Description	Current administrative unit corresponding to the location.
Type	Text (str)
Unit	-
Source	Derived from cross-referencing administrative maps from before

	and after the 2025 reform. https://bando.tphcm.gov.vn/
Min / Max values	-
Date	2026
Notes	Useful when sources use former districts, renamed wards, communes or administrative units changed after the 2025 reform.

Alias name : Water_height_min_cm

Description	Minimum reported water height in centimetres.
Type	Integer (Long)
Unit	cm
Source	Newspapers, Technical reports
Min / Max values	0 ; 50+
Date	2026
Notes	-

Alias name : Water_height_max_cm

Description	Maximum reported water height in centimetres when the source gives a range or a single value.
Type	Integer (Long)
Unit	cm
Source	Newspapers, Technical reports
Min / Max values	0 ; 50+
Date	2026
Notes	For example, for “20-50 cm”, the value recorded here is 50.

Alias name : Depth_class

Description	When the source pro-vides a range, the class is as-signed using the upper value of the range.
Type	Text (str)
Unit	-
Source	Derived from Water_height_max_cm

Min / Max values	-
Date	2026
Notes	Possible values: - Low: h ≤ 15 cm - Moderate: 15 < h ≤ 30 cm - Heavy: 30 < h ≤ 50 cm - Severe: h > 50 cm - Unknown

Alias name : Cause

Description	Main reported or interpreted cause of flooding.
Type	Text (str)
Unit	-
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	Possible values: - Rain - High tide - Combined

Alias name : Impact_type

Description	Main type of impact reported by the source.
Type	Text (str)
Unit	-
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	Possible values: - Traffic disruption - Flooded strets - Infrastructure - Unknown

Alias name : Source_id

Description	Identifier of the source from which the observation was extracted.
Type	Text (str)
Unit	-
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	Example : SRC_1, SRC_2, etc.

Alias name : Source_type

Description	Type of source used for the observation.
Type	Text (str)
Unit	-
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	Possible values: - Newspapers - Technical report

Alias name : Spatial_precision

Description	Qualitative assessment of the spatial precision of the observation.
Type	Text (str)
Unit	
Source	Newspapers, Technical reports
Min / Max values	-
Date	2026
Notes	Possible values: - High (0-100 m) - Medium (100-500 m) - Low (>500 m)

Alias name : Geocoding_method

Description	Method used to place the point geometry from the source. This field documents the geocoding choice made during data integration. It does not describe flood severity.
Type	Text (str)
Unit	-
Source	Derived during data integration
Min / Max values	-
Date	2026
Notes	Possible values: - Exact point - Intersection point - Street point - Area point, etc.

Alias name : Duration_h

Description	Reported duration of the flood observation, expressed in hours when the source provides this information.
Type	Integer (Long)
Unit	Hours
Source	Original sources
Min / Max values	0 ; -
Date	2026
Notes	

Alias name : Observation_notes

Description	Notes to keep methodological comments, uncertainties, translation details, or precision information from the source.
Type	Text (str)
Unit	-
Source	Data integration notes / original sources
Min / Max values	-
Date	2026
Notes	This field is especially useful for explaining approximate water heights, ambiguous administrative names, or choices made during geocoding and classification.